

PROJECT FINAL REPORT

Research Question: Is higher PM2.5 exposure associated with lower life expectancy, and does the Human Development Index (HDI) moderate this relationship?

Motivation

The reason that I chose this project was because air pollution is a problem that affects many people's lives around the world. I wanted to investigate how much air pollution correlates with life expectancy and whether being a developed country (or having a higher HDI) would minimize the effects of the air pollution.

Data Source

I downloaded the data of PM2.5 exposure from World Bank dataset and Life expectancy and HDI index from Our World in Data datasets. I collected the data from these websites in csv form. PM2.5 exposure data has 217 samples of different countries, life expectancy data has 201 samples from different countries and HDI index has 192 data from different countries. All data is from 2020.

Data Analysis

The analysis proceeded in five stages:

1. Data preparation: The three raw CSVs were merged on the ISO3 country code (more reliable than country names) to produce a single dataset of 190 countries for 2020.

2. Exploratory analysis: Distributions, box plots, scatter plots, and correlation matrices revealed that PM2.5 is right-skewed and that PM2.5 and life expectancy move in opposite directions globally.

Figure 1 — Distributions across the 190-country sample (2020)

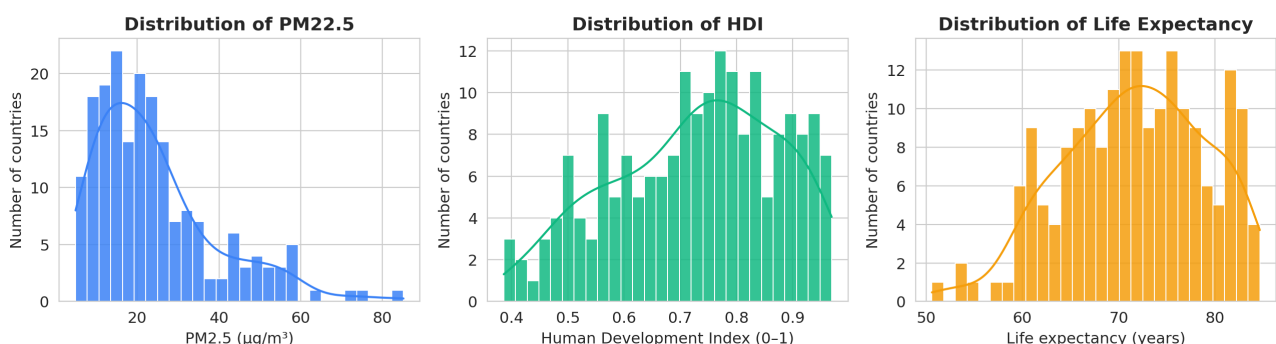


Figure 1. Distributions of PM2.5, HDI, and Life Expectancy across the 190-country sample. PM2.5 is strongly right-skewed; HDI is roughly bimodal; life expectancy is left-skewed.

3. Hypothesis testing: Pearson and Spearman correlations, one-way ANOVA across HDI groups and regions, pairwise t-tests, and a chi-square test of PM2.5 category $\times$ HDI tier — all at $\alpha = 0.05$. A sensitivity analysis re-ran every test with the six Gulf states excluded.

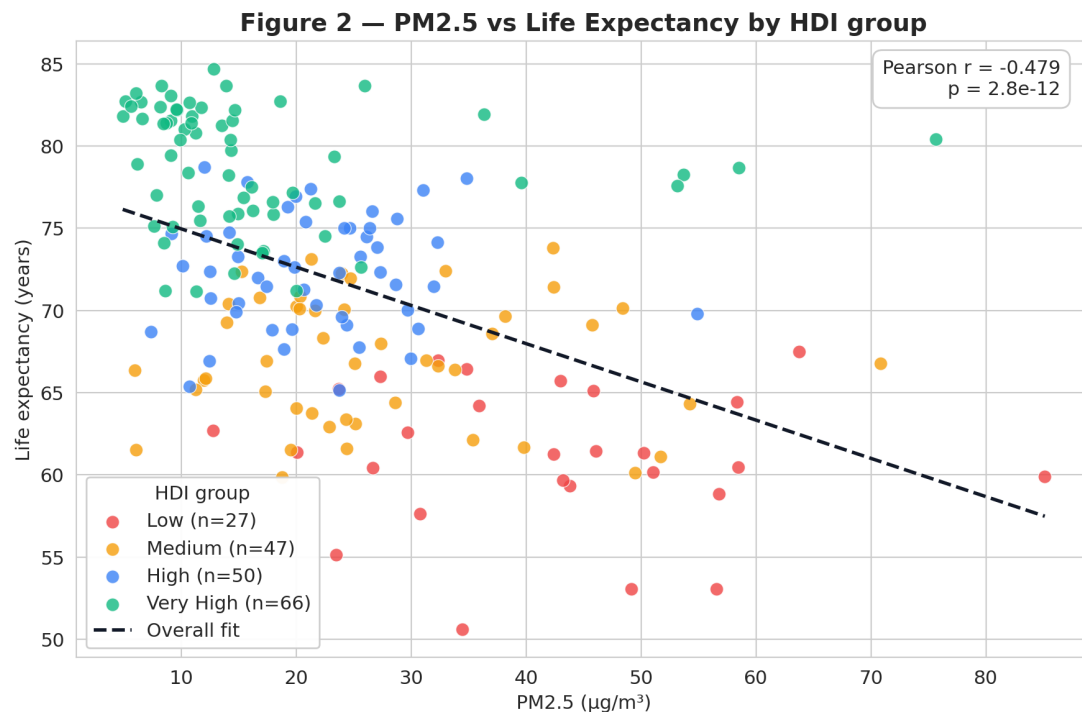


Figure 2. PM2.5 vs Life Expectancy across the 190 countries, colored by HDI group. The strong overall negative correlation (Pearson $r = -0.48$) is visible, but each HDI tier sits in a different cluster — already hinting that HDI is doing most of the work.

4. Linear regression: Three nested OLS models tested confounding versus moderation directly:

M1: LifeExp ~ PM2.5

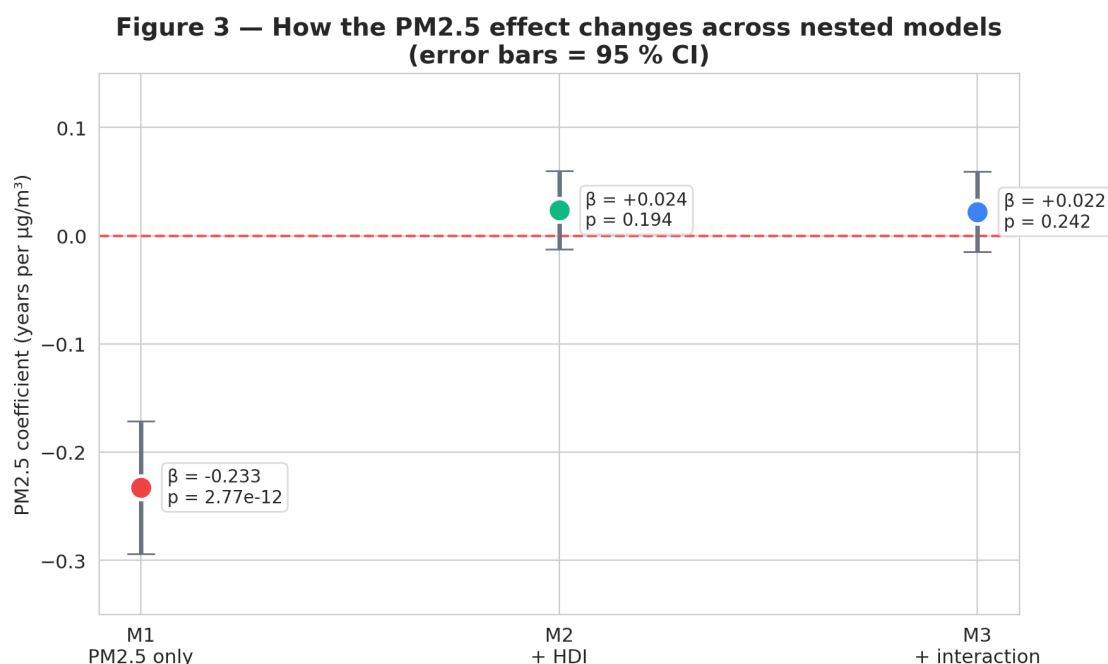
M2: LifeExp ~ PM2.5 + HDI

M3: LifeExp ~ PM2.5 + HDI + (PM2.5 $\times$ HDI)

PM2.5 and HDI were mean-centered in M3. Each model reports coefficients, 95 % CIs, R^2 , adjusted R^2 , RMSE, and 5-fold cross-validated R^2 .

5. Machine learning: Random Forest (with permutation feature importance), K-Means clustering (with elbow + silhouette analysis), and PCA were used as non-parametric and unsupervised checks on the regression results.

Findings



Figure

3. The PM2.5 coefficient across the three nested regression models, with 95 % confidence intervals. In M1, PM2.5 has a strongly significant negative effect ($\beta = -0.23$, $p \approx 10^{-12}$). Once HDI is added in M2, the coefficient collapses to essentially zero and loses significance — the signature of confounding. The interaction term in M3 is also not significant, so there is no evidence of moderation either.

- PM2.5 and life expectancy are negatively correlated globally (Pearson $r = -0.48$, $p \approx 3 \times 10^{-12}$).
- HDI dominates life expectancy ($r = +0.91$). Adding HDI lifts the regression R^2 from 0.23 to 0.82.
- The PM2.5 effect vanishes once HDI is controlled for — its coefficient drops from -0.23 ($p \approx 10^{-12}$) in M1 to $+0.02$ ($p \approx 0.19$) in M2. This is classic confounding.
- No evidence of moderation by HDI. The PM2.5 $\times$ HDI interaction in M3 is not significant ($p \approx 0.69$), contradicting the project's original hypothesis

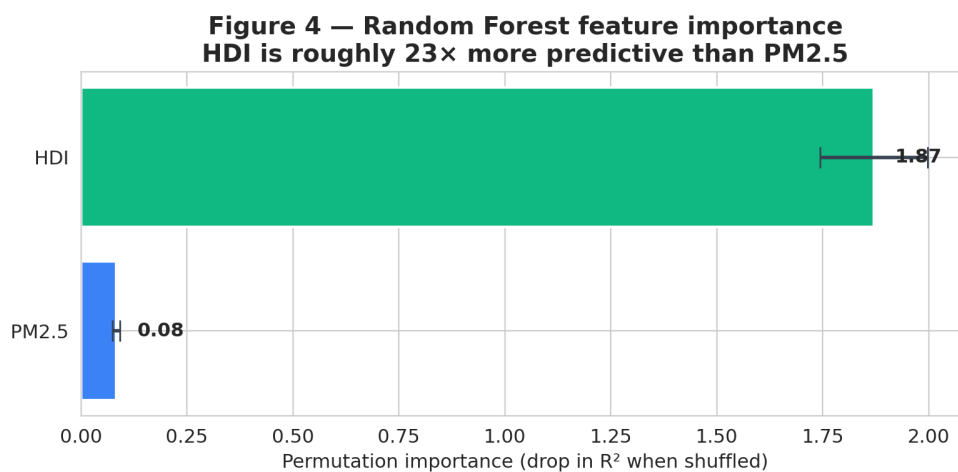


Figure 4. Permutation feature importance from a 500-tree Random Forest. HDI is roughly 23× more predictive of life expectancy than PM2.5, independently confirming the linear-regression conclusion.

- Random Forest agrees. Permutation importance ranks HDI ~23× above PM2.5; the CV R^2 is 0.76, slightly lower than the linear model — so there are no hidden non-linearities.
- Two natural country clusters (silhouette peak at $k = 2$): a "developed / clean / long-lived" group and a "developing / polluted / shorter-lived" group.

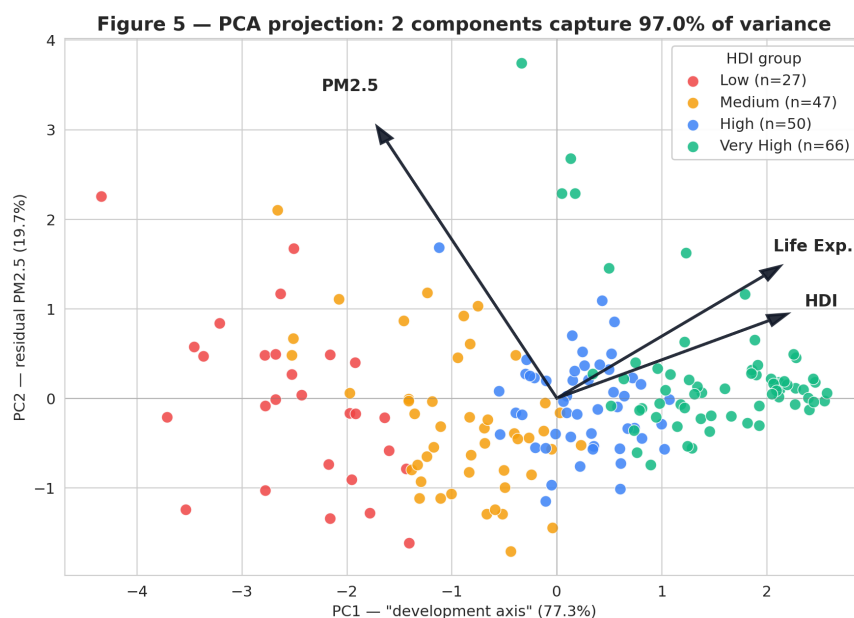


Figure 5. PCA projection of the 3 standardized features into 2D, with loading arrows. Two components capture 97 % of the variance. PC1 (77 %) is a "development axis" along which HDI, life expectancy, and (inverse) PM2.5 all co-vary; PC2 (20 %) is dominated by PM2.5 alone. Countries spread along PC1 in clean HDI-group order.

- PCA captures 97 % of variance in 2 components. PC1 is essentially a "development axis" (HDI and life expectancy load positively; PM2.5 loads negatively); PC2 is residual PM2.5.
- Sensitivity check. Excluding the six Gulf states (whose PM2.5 is natural desert dust) strengthens the within-Very-High-HDI bivariate correlation (r goes from -0.12 to -0.36 , p from 0.35 to 0.004) but does not change the multivariate conclusion that HDI accounts for almost all of the PM2.5 effect.

Overall:

The apparent global relationship between PM2.5 and life expectancy is largely a confounding effect of HDI, not a moderating one. Developed countries have both cleaner air and healthier populations. HDI alone explains about 82 % of cross-country variation in life expectancy, and PM2.5 adds essentially nothing once HDI is in the model. This contradicts the my original moderation hypothesis but is itself a strong, multi-method, internally consistent finding: hypothesis tests, linear regression, Random Forest, and unsupervised methods all point to the same answer.

Limitations and Future Work

Limitations

- Single year (2020) and partly distorted by COVID-era mortality in developed countries.
- Observational, country-level data cannot establish causation. The analysis shows associations and pinpoints confounding; it cannot prove that PM2.5 has no causal effect on health (epidemiological evidence using individual-level data does support such an effect — that evidence is just invisible at this scale).
- Limited feature set: Only PM2.5 and HDI are used as predictors; healthcare spending, smoking, sanitation, and conflict are omitted.
- Small effective sample ($n \approx 65$ in the Very-High-HDI group) limits the power to detect subtle moderation effects.

Future work

- Multi-year panel (2000–2024) to allow within-country fixed-effects and lagged-exposure designs.
- Disaggregate PM2.5 by source (combustion vs natural dust) to separate anthropogenic from natural exposure.
- Sub-national analysis at the city or region level for countries with available data.
- Richer predictor set add healthcare spending, smoking, sanitation, and conflict variables.
- Use HDI's non-life-expectancy components (education + income) to remove the mechanical overlap.
- Causal-inference methods (instrumental variables, regression discontinuity around air-quality policy thresholds) to move from associative to causal claims.

AI Usage

- Claude was utilized as an AI tool in this project.
- Coding support for Streamlit app and pandas
- Help with suitable statistical method selection